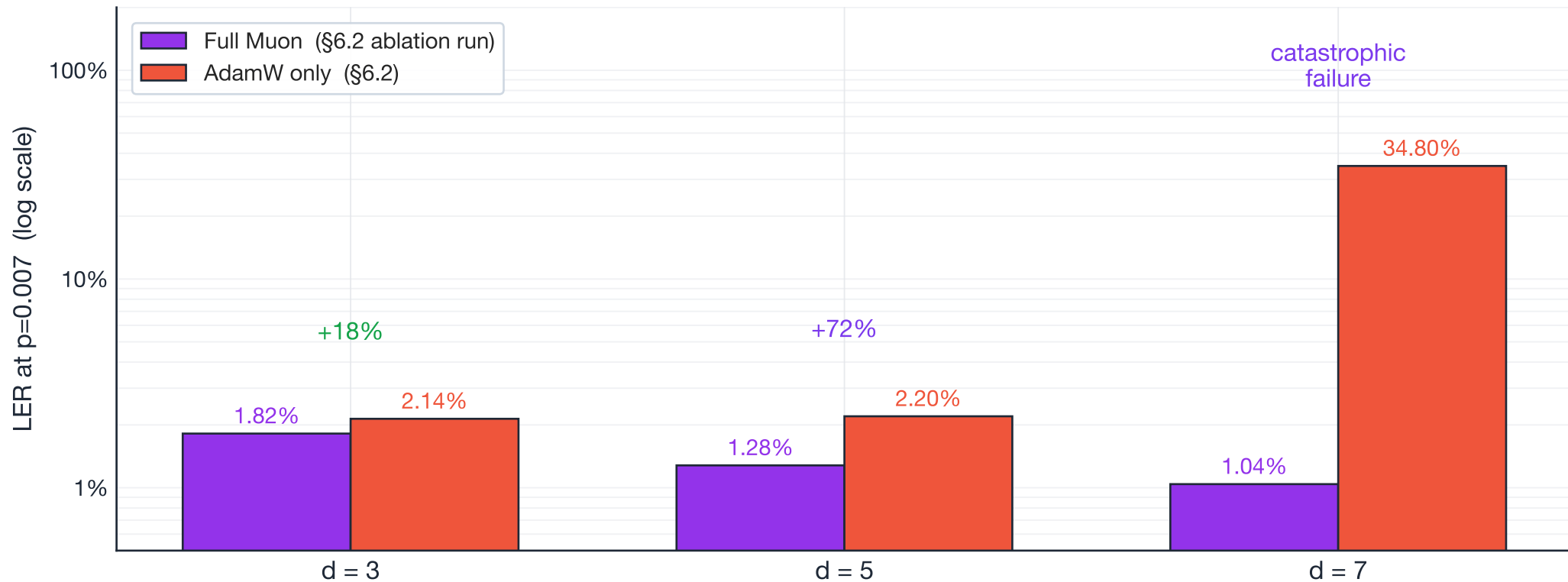


Muon's effect grows with depth — AdamW-only fails to converge at d=7 (within the matched 80K-step budget)



80K training steps / configuration, matched budget (AdamW not separately LR-tuned) — a training-choice comparison, not a tuned-optimizer duel (§4.2/§6.2). AdamW-only $d=7$ fails to escape its initial plateau.